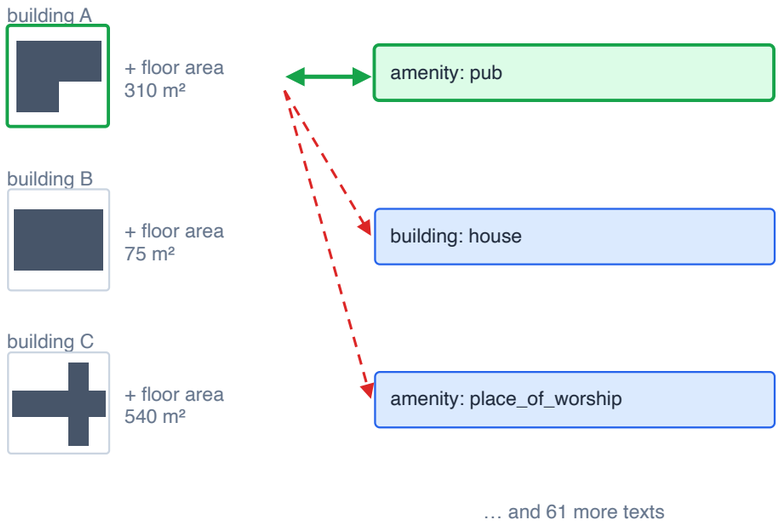
Stage 2: how CityFM learns building types from footprints, and how we test it

Each building is represented twice: by its outline and floor area, and by its tags read as text without names (e.g. *amenity: pub*). Stage 2 trains only the footprint side.

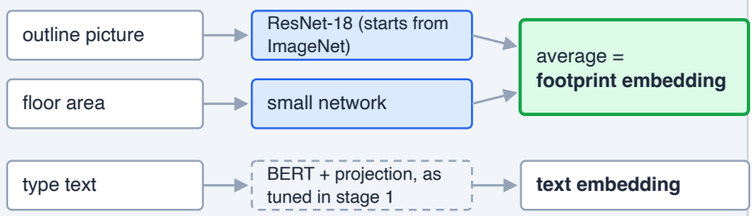
the right type text · a footprint picture + floor area · made more similar: its own type text · made less similar: the other 63 texts and footprints · kept fixed (not trained in stage 2)

1 Training

Take 64 buildings whose type texts all differ. The model is adjusted so that each footprint embedding is more similar to its own type text than to the other 63 texts and 63 footprints. Only the footprint side is trained.



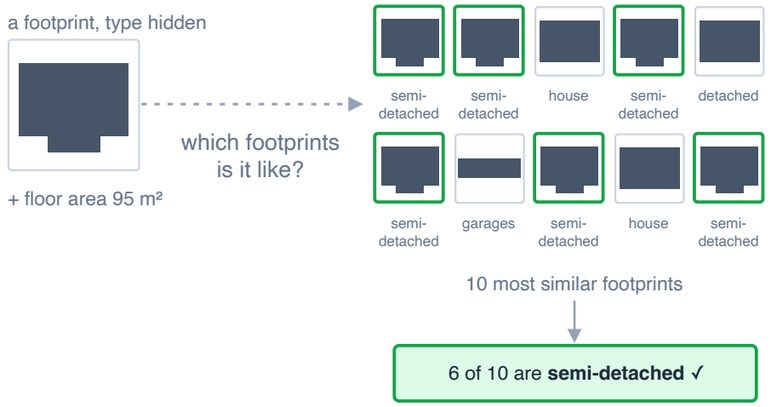
How each side becomes 768 numbers



Floor area is capped at 5,000 m². Pictures are 224 × 224, drawn in metres.

2 Test 1 · Footprint compared with footprints

18 common types, up to 300 of each. The type is predicted from the footprint embedding in two ways: from its 10 most similar footprints, or with a trained classifier.



10 most similar footprints: right type, out of 18

Chance	5.6%
Floor area alone	19%
Untrained image network	19%
CityFM	**57%**

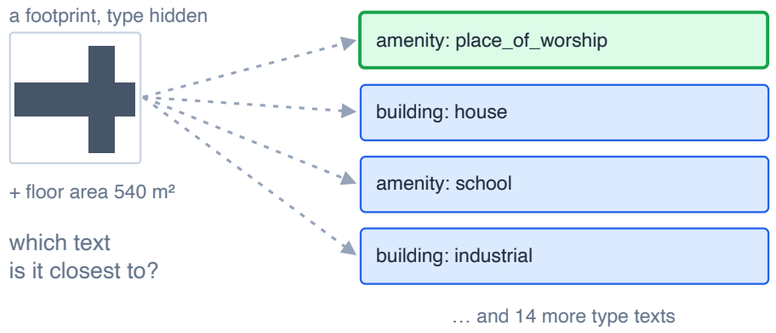
Trained classifier: right type, out of 18

Chance	5.6%
Floor area alone	21%
Untrained network + area	27%
CityFM (+ area)	**57%**

Same 57% either way. Most misses are between look-alikes: house types and garages.

3 Test 2 · Footprint compared with type texts

Nothing is trained for this test. Each footprint is assigned the type whose text embedding is most similar to its footprint embedding, out of the 18 types.



Most similar type text: right type, out of 18

Chance	5.6%
CityFM	**47%**

No baselines here: area alone and the untrained network have no text embeddings to compare with. Best matched: pubs, places of worship, schools.

Training quiz: own text first, out of 64

Chance	1.6%
CityFM	**43%**

Not a held-out test. The quiz draws from the same buildings training used, and training saw about 320,000 footprints, more than Leeds has (262,714). Most quiz buildings were seen in training, so 43% is close to a training score.

What it shows. After stage 2, the footprint embedding distinguishes building types by shape and size. It gives the right type 57% of the time with either method, against 19-27% from floor area or the untrained image network. The type text most similar to a footprint is its own type 47% of the time, with no training for this test. The remaining errors are mostly between types that look alike or mean much the same: house types, *building: school* and *amenity: school*, farms and farm buildings.

Chance: 1 in 18 types (1 in 64 in the quiz). **Floor area alone**: no picture. **Untrained image network**: ResNet-18 with its ImageNet weights, before stage 2 (outline only; area added in the classifier). 5,269 buildings; same scale in all charts.